

CreditPay

Credit Card Payment System – Sprint Submission

Submitted by	Aprajit Roshan
Submission date	08 October 2026
Test account	testuser (testuser@example.com)
Currency	Indian Rupee (₹)

1. Sprint Overview

This document summarises the work delivered for the sprint and provides screenshot evidence for each item. The sprint covered four features: automated email notifications, dark mode, monthly statement PDF generation, and an admin panel for card management.

#	Sprint item	Evidence in this document
1	Automated email notifications (high-value transaction, card blocked, low available credit)	Section 2 – three email alerts received
2	Dark mode using React Context API	Section 3 – dashboard in dark mode with theme toggle
3	Monthly statement PDF generation	Section 4 – requirement checklist
4	Admin panel for card management	Section 5 – admin dashboard, card management, card activity

2. Automated Email Notification System

The system sends an email alert automatically when any of three conditions is met. Each alert below was received in the Gmail inbox of the test account.

2.1 Transaction amount exceeds ₹5,000

A payment of ₹6,000.00 triggered the “Credit Card Payment Alert” email, received at 14:00. The email states the transaction amount and the ₹5,000 threshold that was exceeded.

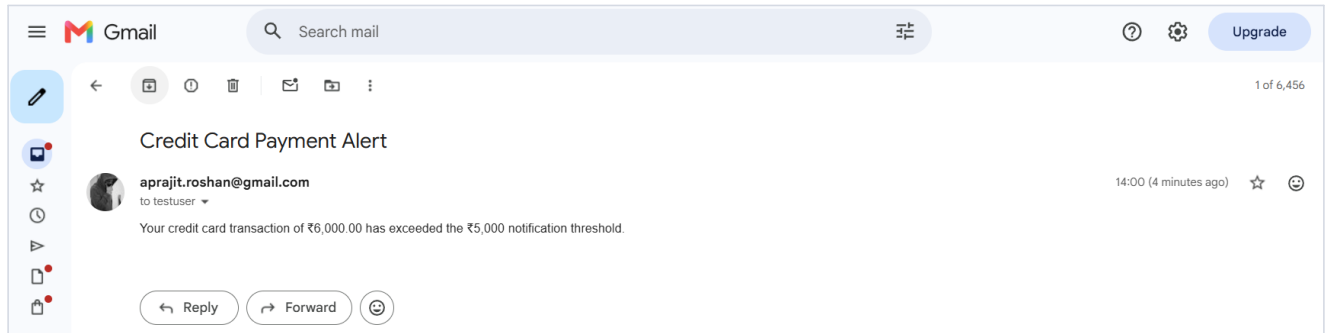


Figure 1 – “Credit Card Payment Alert” email (₹6,000.00 transaction vs ₹5,000 threshold)

2.2 Card is blocked

When an administrator blocked the card, a “Credit Card Blocked” email was sent at 14:54. The card number is masked and only the last four digits (4282) are shown.



Figure 2 – “Credit Card Blocked” email (card ending 4282)

2.3 Available credit falls below 10%

After spending, the available credit dropped to ₹5,000 of a ₹75,000 limit (6.67%), below the 10% threshold, and a “Low Available Credit Alert” was sent at 17:38 showing the limit, available credit and percentage.

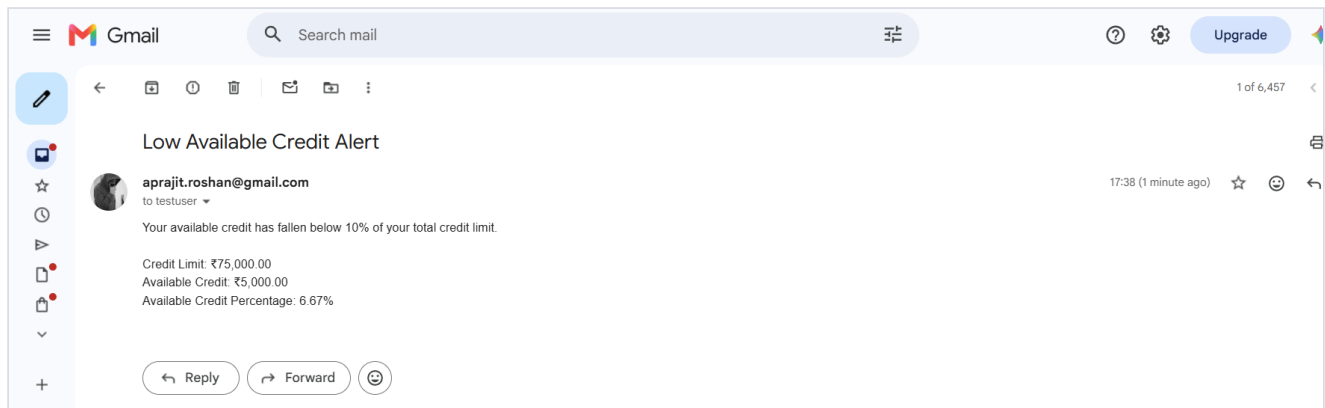


Figure 3 – “Low Available Credit Alert” email (₹5,000 of ₹75,000 = 6.67%)

Summary of notification triggers

Trigger	Condition	Email subject	Result
High-value transaction	Amount > ₹5,000	Credit Card Payment Alert	Verified (Fig. 1)
Card blocked	Card status set to blocked	Credit Card Blocked	Verified (Fig. 2)
Low available credit	Available credit < 10% of limit	Low Available Credit Alert	Verified (Fig. 3)

Note: card numbers in the notifications are masked, showing only the last four digits.

3. Dark Mode Support

The dashboard UI supports a dark theme managed through a React Context provider, so any component can read and toggle the theme without prop drilling. A theme toggle button is available in the top-right corner of every page, including the admin pages.

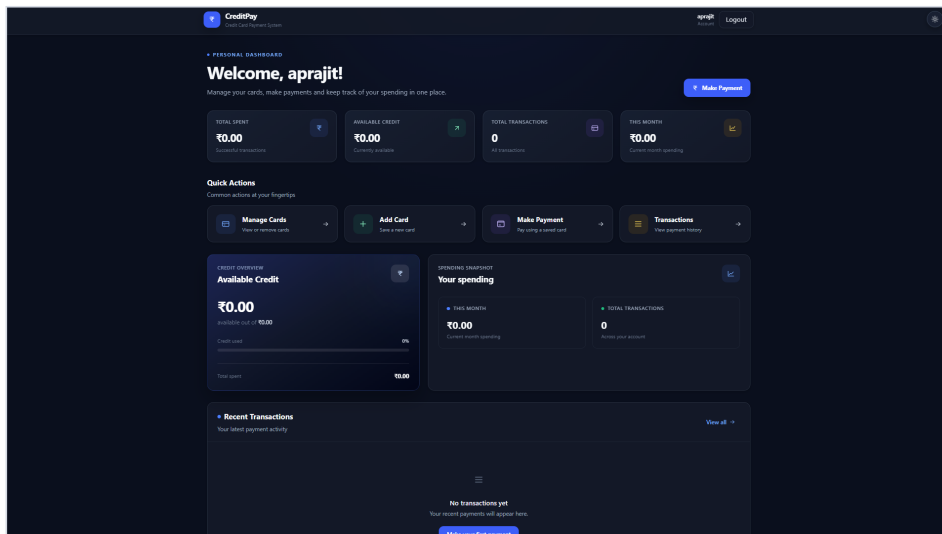


Figure 4 – User dashboard in dark mode (theme toggle at top right)

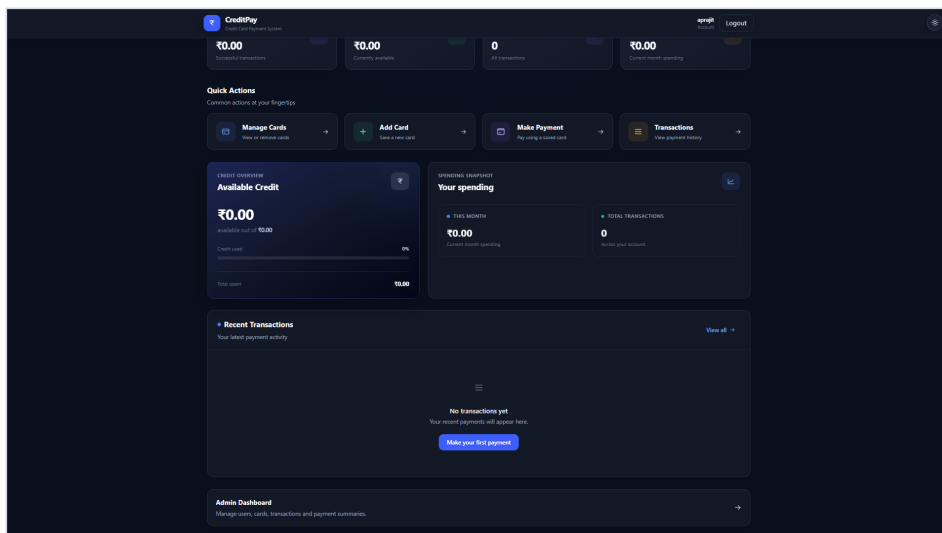


Figure 5 – Scrolled dashboard showing consistent theming and the Admin Dashboard entry point

Requirement coverage

Requirement	Status / evidence
Theme switching between light and dark	Toggle button present in header on all pages (Figs. 4-8)
Theme persistence across sessions	Required: preference persisted and restored on reload
Responsive and accessible design	Readable contrast in dark mode; status shown with text labels as well as colour (Active / Success / Failed)

4. Monthly Statement PDF Generation

Per the sprint scope, a backend endpoint generates a downloadable monthly statement PDF in a clean, professional financial layout.

Statement contents (per scope)

Section	Details
Header	Account holder, statement month and generation date
Masked card details	Card number masked, showing only the last four digits (e.g. *****4282)
Transaction list	Date, payment reference, amount and status for each transaction in the month
Total spending	Sum of successful transactions for the month
Summary information	Credit limit, available credit and transaction counts

Sample statement output can be attached as an appendix to this document.

5. Admin Panel for Card Management

The admin area is reached through the Admin Dashboard entry on the main dashboard (Fig. 5).

5.1 Admin dashboard

The admin dashboard summarises system activity: 4 users, 2 cards, 10 transactions (5 successful, 5 failed, 0 pending) and a total successful payment amount of ₹10,000.00. The daily summary for 2026-10-08 shows 1 successful payment (₹6,000.00) and 1 failed payment. Card Management and a CSV Transaction Export are available as management tools.

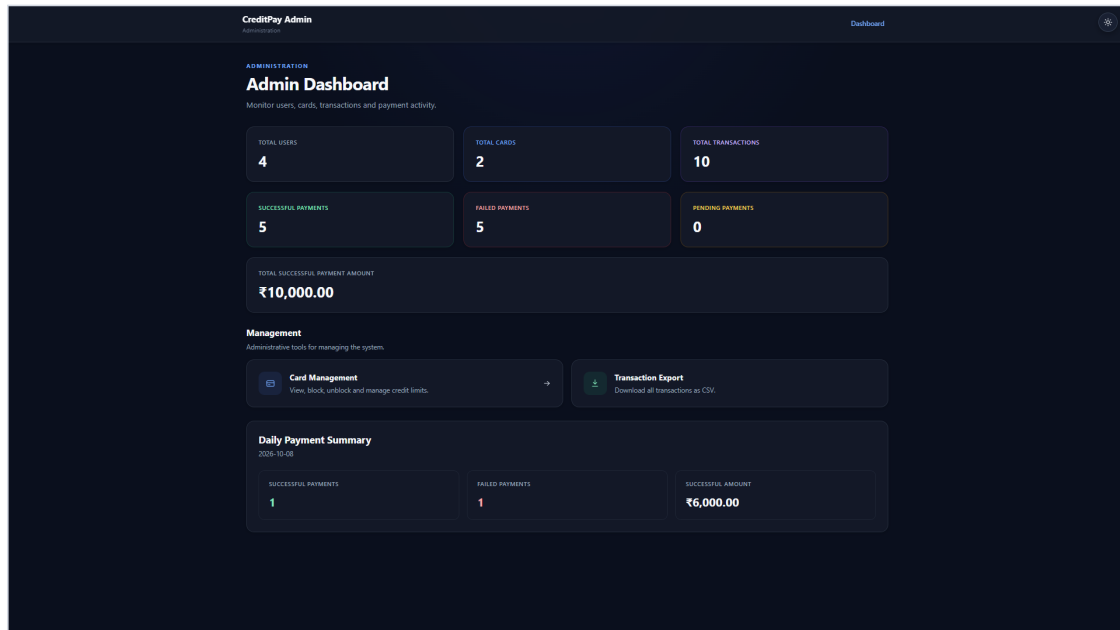


Figure 6 – Admin dashboard with totals, management tools and daily payment summary

5.2 Card management

Administrators can view all customer cards with holder name, account, card type, expiry and credit limit, and take action on each card: block (or unblock) the card, set a new credit limit, and open the activity view. Counters at the top show total, active and blocked cards.

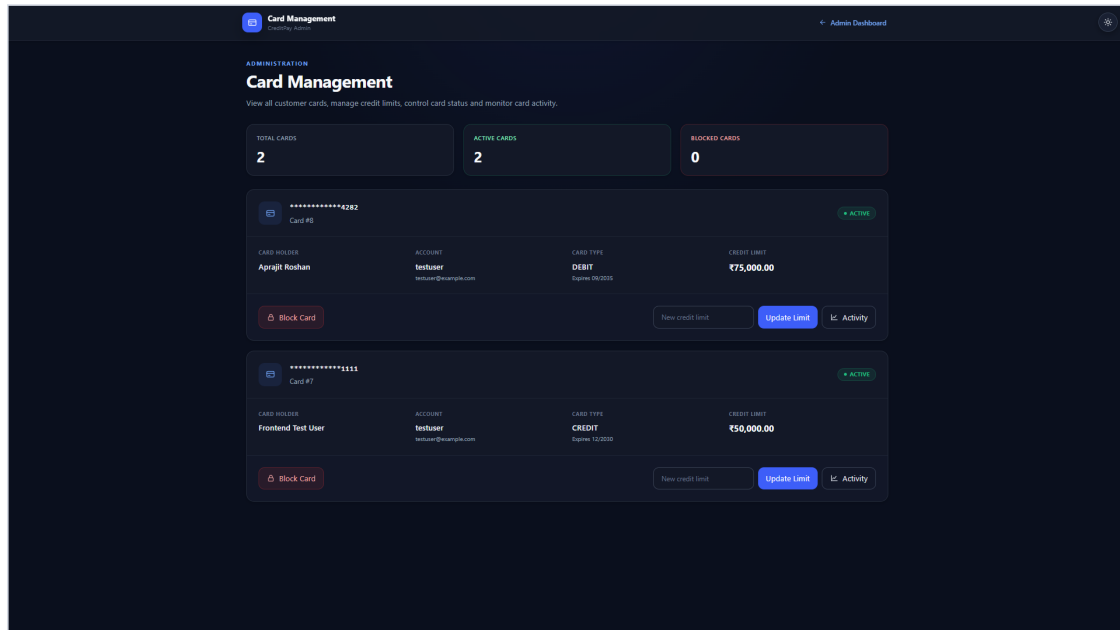


Figure 7 – Card Management: all cards with Block Card, Update Limit and Activity controls

5.3 Card activity monitoring

The Activity view lists the payments made on a card with amount, payment reference, timestamp and status. For card ending 4282 it shows Payment #14 (₹6,500.00, Failed), Payment #13 (₹6,000.00, Success) and Payment #12 (₹2,000.00, Success).

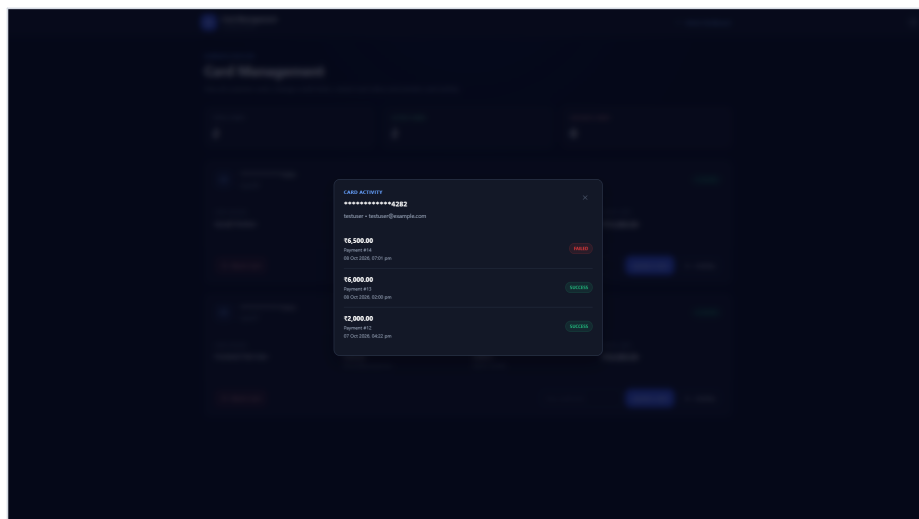


Figure 8 – Card Activity modal for card ending 4282

Requirement coverage

Requirement	Status / evidence
View all card details	Fig. 7 – holder, account, type, expiry, credit limit, status
Block and unblock cards	Block Card control (Fig. 7); block email sent (Fig. 2)
Update credit limits	Update Limit control with new-limit input (Fig. 7)
Monitor card activity	Activity modal (Fig. 8) and daily summary (Fig. 6)
Proper access controls	Separate admin area, reached via the Admin Dashboard entry (Fig. 5)

6. Conclusion

The sprint items are covered as follows: notification emails were received for each trigger, the UI runs in dark mode with a theme toggle, the monthly statement feature is specified above, and administrators can manage cards and monitor activity from the admin panel.